

Spain Living Costs

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Project Overview

I am building an Interactive Data-Driven Recommendation Model to analyze which cities in Spain provide the best combination of affordability, employment opportunities, and quality of life.

The Problem

The cost of living and quality of life can vary significantly between cities in Spain. A city with higher salaries may also have significantly higher housing costs, while a smaller city may be much more affordable but offer fewer employment opportunities.

My goal is to determine which Spanish cities provide the best overall value based on factors such as income, rent, housing prices, employment, transportation, safety, and lifestyle. Instead of having one fixed score, I will develop a model that allows users to adjust their personal priorities and receive a personalized ranking of cities.

Tools

I currently plan to use several tools:

- Python: Data cleaning, scoring model, interactive model
- Excel: Initial data exploration, organization

Data

The primary data source will be the Spanish National Statistics Institute Urban Indicators. This dataset provides city-level information including:

- average annual incomes
- rent and housing prices
- employment and unemployment
- population and demographics
- transportation
- tourism
- safety/crime indicators

I will also use the INE Household Budget Survey to provide information about typical spending patterns, including housing, food, transportation, restaurants, recreation, and other household expenses

Deliverables

The final product will be an interactive dashboard featuring:

- an interactive map of Spain

- city rankings
- city-to-city comparisons
- individual city profiles
- interactive charts and visualizations
- recommendation based on user income and preferences

Weekly Timeline

Week	Project Progress
1	Finalize research question, define variables, collect initial INE data
2	Clean and organize the datasets, handle missing values and combine variables
3	Conduct exploratory data analysis and examine income, rent, housing burden, and employment patterns
4	Create initial visualizations and investigate relationships between major variables
5	Develop affordability, housing, employment, transportation, lifestyle, and safety scores
6	Build and test the overall city ranking and personalized weighting system
7	Conduct advanced analysis
8	Begin building Streamlit dashboard
9	Add personalized recommendation system using income and user priorities
10	Add interactive charts, city comparisons, profiles
11	Test the model, validate results, fix errors, create example profiles
12	Finalize dashboard, prepare presentation